

QUANTUM SAFE GATEWAY

1. Project Overview

Quantum Safe Gateway is a cybersecurity project focused on protecting network communication against future quantum-computing threats.

The project uses a secure gateway architecture with TLS 1.3, Mutual TLS, X25519 and ML-KEM to establish secure communication between clients and backend services.

Goal: To explore a modern security gateway that combines existing cryptographic methods with post-quantum cryptography.

2. How It Works

Client → Secure Gateway → Authentication & Encryption → Backend Service

- Client connects securely to the gateway.
- Mutual TLS is used for authentication.
- Cryptographic key exchange is performed using modern security mechanisms.
- X25519 + ML-KEM are used to explore hybrid/classical + post-quantum security.
- The gateway securely forwards communication to the protected backend.

3. Technologies Used

Go (Golang) · TLS 1.3 · mTLS · X25519 · ML-KEM · Post-Quantum Cryptography · Secure Networking

4. Key Outcome

The project provided practical experience in:

- Secure network communication
- Cryptographic key exchange
- TLS and mutual authentication
- Post-quantum security concepts
- Backend development using Go

5. Applications

- Secure APIs
- Enterprise network gateways
- Cloud services
- Financial systems
- Future quantum-resistant infrastructure

6. Future Scope

The gateway can be further developed with additional post-quantum algorithms, performance optimization, monitoring, automated certificate management, and cloud deployment.

APPENDIX



Architecture

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2026/04/06 13:51:46 Gateway running on https://localhost:8043
2026/04/06 13:52:16 crypto_policy_reloaded hash=08308c52c71bd701e6c5596388d4341776ab916bee92009ff6e0854a07d7ae37
2026/04/06 13:52:17 ---- AUTH MIDDLEWARE START ----
2026/04/06 13:52:17 ❌ Missing Authorization header
2026/04/06 13:52:46 crypto_policy_reloaded hash=08308c52c71bd701e6c5596388d4341776ab916bee92009ff6e0854a07d7ae37
2026/04/06 13:52:53 ---- AUTH MIDDLEWARE START ----
2026/04/06 13:52:53 ❌ Payload JSON parse failed: invalid character 'c' looking for beginning of value
2026/04/06 13:53:16 crypto_policy_reloaded hash=08308c52c71bd701e6c5596388d4341776ab916bee92009ff6e0854a07d7ae37
2026/04/06 13:53:26 ---- AUTH MIDDLEWARE START ----
2026/04/06 13:53:26 🟢 Token parsed successfully
2026/04/06 13:53:26 SUB: 123
2026/04/06 13:53:26 EXP: 4700000000
2026/04/06 13:53:26 ---- AUTH MIDDLEWARE PASSED ----
2026/04/06 13:53:26 =====
2026/04/06 13:53:26 NEW REQUEST RECEIVED
2026/04/06 13:53:26 Time: 2026-04-06 13:53:26
2026/04/06 13:53:26 Client IP: [::1]:63294
2026/04/06 13:53:26 Method: GET
2026/04/06 13:53:26 Path: /
2026/04/06 13:53:26 Token Subject: 123
2026/04/06 13:53:26 Token Expiry: 4700000000
2026/04/06 13:53:26 Token VALID (within skew)
2026/04/06 13:53:26 TLS Version: 772
2026/04/06 13:53:26 Cipher Suite: 4885
2026/04/06 13:53:26 Client Ch: legacy-client
2026/04/06 13:53:26 Client Cert SHA256: 9be6553fadd358bb61f4372e6264f894531d6142cd5cc402e7789ecf9bec8b
2026/04/06 13:53:26 ----- POLICY APPLIED -----
2026/04/06 13:53:26 Client: legacy-client
2026/04/06 13:53:26 Policy Classical: true
2026/04/06 13:53:26 Policy Cipher Suites: [TLS_AES_256_GCM_SHA384]
2026/04/06 13:53:26 -----
2026/04/06 13:53:26 PQC Shared Secret Match
2026/04/06 13:53:26 Audit Status: SUCCESS
2026/04/06 13:53:26 RESPONSE SENT
2026/04/06 13:53:26 =====

```

Implementation